Patient-Wise Splitting Degrades Into Random Splitting as Participant Identifiers Fail: A Controlled Measurement in Alzheimer's Disease MRI

Mehmet Berke Isler[1,2*], Irem Ilter[3] and Mehmet Kemal Ozdemir[2]

[1*]International School of Medicine, Istanbul Medipol University, Istanbul, Türkiye.
[2]Department of Computer Engineering, School of Engineering and Natural Sciences, Istanbul Medipol University, Istanbul, Türkiye.
[3]Institute of Neuroinformatics, ETH Zurich and University of Zurich, Zurich, Switzerland.

*Corresponding author(s). E-mail(s): mehmetberkeisler@gmail.com;
Contributing authors: ilteri@student.ethz.ch; mkozdemir@medipol.edu.tr;

Abstract

Patient-wise splitting is the standard remedy for leakage in medical imaging, and its guarantee is usually credited to the splitting procedure. It is not a property of the procedure. It is a property of the participant identifier the procedure is given, and identifiers are deleted, fragmented and merged by redistribution. We measure that dependence across its full range on ADNI1 (**220** cognitively normal and AD participants, **1,123** scans, one 2D coronal-centre slice per scan). With identifiers complete, subject-wise splitting reports an area under the receiver operating characteristic curve (AUROC) of **0.771**; with every identifier removed the same unchanged procedure reports **0.946**, against **0.947** for random image-level splitting, an inflation of **+0.176** accrued without any error being raised. A provenance-aware grouping adds nothing while identifiers are intact, which is the design target, and declines optimism in **7** of **8** degraded conditions (Wilcoxon $p = 0.008$). Permuting diagnosis across participants removes the disease signal, yet random splitting still reaches **0.878** AUROC while both patient-aware protocols sit at chance, identifying patient recall as the mechanism. One instance is observed rather than imposed: a widely used public 2D AD benchmark is, by content matching, a redistribution of **200** OASIS-1 participants, and grouping by the only identifier it supplies still leaves **70%** of test images sharing a

participant with training. Manifests, audit code and the rebuild pipeline are released under Apache-2.0.

Keywords: Patient-identity leakage, Evaluation protocol, Alzheimer's disease, Structural MRI, Data provenance, Reproducibility

1 Introduction

Deep-learning models for Alzheimer's disease (AD) classification from structural MRI routinely report near-perfect benchmark performance that does not survive patient-independent evaluation [1–3]. The accepted remedy is to split by patient, and the guarantee that follows is usually credited to the splitting procedure.

Patient identity is one shortcut among several that a model can take instead of learning pathology [4–6], and it is the one a split is supposed to close. Leakage of this kind is a recognised contributor to the reproducibility problem in applied machine learning [7–10].

It is not a property of the procedure. A procedure can only keep a patient's images together if it is told which images belong to that patient, and participant identifiers are deleted, fragmented and merged whenever a cohort is redistributed. Patient-wise splitting is therefore a property of the identifier the procedure is given, and the strength of the guarantee varies with the quality of that evidence.

The two ends of this axis are established. Random image-level splitting is known to inflate, on individual AD cohorts [1, 2], in longitudinal brain MRI [11], and in connectome models [12]; patient-wise splitting is known to correct it, and tools exist to enforce it [13]. What lies between has not been measured. That is the interval redistributed benchmarks occupy, and §3.5 shows one of them sitting in it.

Two concurrent audits reach that interval from the other direction, by finding benchmarks already there. Sifat *et al.* [14] trace subject identifiers from the slice captions of a Kaggle redistribution of OASIS-1 and report that its subject count does not match the labelled source cohort. Vangala *et al.* [15] recover patient identity for the traceable 12.6% of a public brain-tumour test split, find that 95.5% of that subset shares a participant with training, and report that removing every near-duplicate test image leaves balanced accuracy unchanged. We take both as evidence that the condition is common rather than exotic. Neither measures the path between the endpoints, and neither retrains.

Contributions

First, and principally, we measure the function relating evaluation validity to identifier quality on one cohort in one code state. With identifiers complete, subject-wise splitting reports 0.771 area under the receiver operating characteristic curve (AUROC); with every identifier removed the same unchanged procedure reports 0.946, against 0.947 for the random image-level splitting it exists to prevent, an inflation of +0.176 accrued without the procedure raising an error at any point. Second, we separate patient recall from disease signal with a permutation control, an established instrument [16], and locate the regime in which a provenance-aware grouping helps. Third, we release SplitGuard-AD, an executable audit that builds a per-image leakage graph from whatever identifier evidence a cohort carries. We offer it as the instrument

that makes the measurement possible rather than as an improvement on participant grouping: where identity is intact its component layer reduces to participant grouping, as it should, and the benefit it carries appears only once provenance degrades. Both halves of that statement are reported as results.

2 Materials and Methods

2.1 Cohorts

ADNI1 (Tier 3) is the headline cohort, used under an active LONI data use agreement [17–19]. The collection holds 2,182 pre-processed 1.5T T1-weighted volumes from 382 participants; the CN-versus-AD primary arm is 220 participants and 1,123 diagnosis-resolved scans, with converter components excluded. Each volume is reduced to one 2D coronal-centre slice at ingest. Labels join imaging to the clinical table on `(RID, VISCODE)`, with a 180-day date-proximity fallback for 22.2% of scans; Supplementary Material S5 accounts for that join per scan.

OASIS-1 (Tier 2) is the replication cohort, where participant identifiers are authoritative [20]. It is operationally external to ADNI: different site, scanners, protocols and recruitment era. The direction-preservation we report between the two is a partial cross-site check, not external validation in the regulatory sense.

A redistributed public benchmark (Tier 1) is the observed instance. It is a widely used 2D AD benchmark of 6,400 images in four CDR-derived classes. By content matching against OASIS-1 we recover its complete participant mapping: all 6,400 images resolve unambiguously to 200 OASIS-1 participants at 32 consecutive axial slices each, with the weakest correct match correlating at 0.9996 against a strongest cross-participant match of 0.9623. Supplementary Material S8 gives the mapping audit. Because all 200 participants are also in our Tier-2 set, the cross-tier replication rests on two source cohorts rather than three.

2.2 The leakage graph and the three protocols

SplitGuard-AD builds an undirected graph over images and joins two images whenever any surviving identifier says they belong to one participant: the same subject key, session, series UID, or acquisition date scoped within a participant. Missing values are treated as absent evidence rather than as a shared key, which matters because a literal `unknown` used as an identifier would merge every unlabelled scan into one fictional participant and the audit would then pass the split. Connected components of this graph are the indivisible units, and a class-stratified bin-packer allocates whole components to partitions.

The same reasoning applies to the grouped splitters in general-purpose libraries and medical-imaging frameworks [21, 22]: each takes a group label as input and is only as reliable as that label.

Three protocols are compared on the same image universe. **Protocol A** shuffles images at random and is the leaky protocol. **Protocol B** splits by the supplied subject identity with no component constraint. **Protocol C** is the component-safe split. All three use the same assigner, so a difference between them is attributable to the grouping and not to the assignment step. Where identity is complete, B and C coincide by construction.

2.3 Degrading and corrupting provenance

Provenance integrity is varied in two ways on the ADNI primary arm, at five seeds each. Identifiers are *deleted* per scan at rates $\{0, 10, 25, 50, 100\}\%$, which models metadata loss. They are separately *corrupted* in the two ways redistribution corrupts them: SPLIT deals one participant across several keys, and MERGE places several participants under one, each at intensity $\lambda \in \{0, 0.1, 0.25, 0.5, 1\}$. True identities are retained out of band, so leakage is scored against ground truth rather than against the damaged key. These are controlled stress tests of identifiable failure modes, not estimates of their prevalence in clinical repositories.

Training is held fixed throughout: ResNet-18 with ImageNet initialisation, 15 epochs, AdamW at 10^{-4}, class-balanced loss, seeds $\{0, 1, 2, 3, 4\}$, deterministic kernels and seeded loaders. Supplementary Material S1 gives the full configuration. Intervals are bootstrap intervals [23, 24] over five seeds and are descriptive throughout, not significance tests. Every AUROC reported here comes from one code state, approximately 14.8 GPU-hours in total; the package list frozen on the final node is released as `runs_gpu/gpu_environment.txt`, and `scripts/verify_paper_numbers.py` holds every number in this manuscript to the artefact that produced it.

3 Results

3.1 Reference: the gap under intact provenance

On the ADNI1 primary arm, random image-level splitting reports 0.947 AUROC and component-safe splitting 0.829, an inflation gap of +0.118 (paired-seed interval [+0.096, +0.145], hierarchical [+0.061, +0.181]; $n = 5$, descriptive). The gap survives every prespecified variation of cohort, backbone, converter rule, acquisition type, input dimensionality and label-linkage strictness (Figure 1): +0.146 under DenseNet-121, +0.035 on OASIS-1, +0.235 on a 3D volumetric arm, and +0.113 when the weaker quarter of the label chain is dropped. No arm's interval includes zero.

Decomposed, the whole of that effect is the subject layer. The component marginal, Protocol B minus Protocol C, includes zero in every arm. That is the expected result and the control for everything that follows: with authoritative identifiers a provenance-aware grouping should agree with participant grouping, and it does.

3.2 The mechanism is patient recall

The gap alone cannot separate "the leaky protocol learned the disease better" from "the leaky protocol recognised its patients". We separate them by removing the disease. The instrument is not new: Chaibub Neto *et al.* [16] introduced permutation tests for exactly this purpose, and label-shuffling controls of the same shape are now standard in neurophysiology benchmarks.

Diagnosis labels are permuted across participants, once over the whole cohort so that a participant straddling a boundary carries the same permuted label on both sides, and the three protocols are retrained unchanged. Random splitting still reaches 0.878 AUROC. Subject-only sits at 0.521 and component-safe at 0.427, both chance. The permutation is verified to have taken effect: permuted labels coincide with the true diagnosis 48.3%, 55.1% and 50.3% of the time, which is the chance rate. All three protocols fit the training set equally well; only the random split's memorisation transfers, because only its test set contains the same people.

A model with no pathology available to learn cannot be exploiting pathology. That structural MRI carries recoverable identity is independently established [25–27]; what the control adds is that a classifier trained for diagnosis uses it.

3.3 Evaluation validity as a function of identifier quality

Deleting identifiers at controlled rates traces the function (Figure 2, Table 1). At zero loss the two patient-aware protocols admit no leakage and are indistinguishable, reproducing §3.1. They separate immediately thereafter: at 10% loss the graph prevents 49% of the participant leakage subject-wise splitting admits (9.4 against 18.6 straddling participants), and 30 to 37% at higher rates. The amplification is non-linear. Deleting 10% of identifiers returns 36% of test participants to training under Protocol B, because a participant contributes a median of five scans and one of them losing its index is enough to detach that scan from the group.

Following the AUROC to the end of the sweep gives the result this paper is built on. With identifiers complete, subject-wise splitting reports 0.771 $[0.709, 0.833]$. With every identifier removed, the same procedure, unchanged, reports 0.946 $[0.921, 0.972]$. The random image-level split of §3.1 reports 0.947 $[0.926, 0.965]$ on the same cohort. The difference between the last two is 0.001 and their intervals overlap across almost their whole length.

We state that as a failure to distinguish rather than as an equality: five seeds cannot establish that two procedures agree. What the comparison does establish is that nothing in the reported number separates them. A reader handed the right-hand column cannot tell that the split was meant to be patient-disjoint, and neither can the procedure, which raises no error: every scan is a singleton, every singleton is its own group, and grouping by group is vacuous. The inflation across the sweep is +0.176 AUROC, accrued without a warning at any point.

That is the sense in which patient-wise splitting is not a property of a splitting algorithm. The algorithm is unchanged along this axis. Only the evidence it is given changes, and the guarantee it is credited with degrades continuously into the condition it was adopted to prevent.

3.4 Structured corruption: where the graph earns its place

Deletion is the visible failure. Redistribution corrupts identifiers invisibly, and SPLIT and MERGE model the two directions (Table 2). Under fragmentation the graph recovers most of what the participant key lost, because session and series edges do not take the subject key as an input: at $\lambda = 1$ it prevents 90% of the straddling participants subject-wise splitting admits (6.4 against 61.6). Under merging it prevents nothing, and nothing is the correct answer: over-grouping destroys the participant count without placing any participant on both sides of a boundary, so there is no leakage to prevent.

Across the eight degraded cells where a difference is defined, the component grouping declined optimism the subject-only grouping accepted in seven, with a mean of +0.026 AUROC and a one-sided Wilcoxon $p = 0.008$. The cells share seeds and a base manifest, so this is a consistent direction rather than an independent test, and we report it as such. Taken with §3.1, where the same layer buys nothing on intact data, the pattern is the one the framework predicts: the graph is worth its cost exactly where some identifier other than the participant key survives.

3.5 An instance observed rather than imposed

The deletion sweep is synthetic. Tier 1 is the same condition found in the wild (Table 3). Its filename index is not a participant identifier: 324 distinct keys divide into 149 that match exactly one participant and 175 that merge two or three, while 179 of the 200 participants are spread across two or more keys and one across 32. Grouping by that key, the only identifier the release supplies, leaves 88.6 of test images sharing a participant with training.

What that costs depends on which protocol a user adopts. Random splitting, which is what the published literature on this benchmark does, reports 0.999. Grouping by the filename key reports 0.793, and grouping by the recovered participant identity reports 0.766. The difference between the first and the last is +0.233 [+0.191, +0.270]. The broken key therefore recovers most of the correction while leaving seven test images in ten contaminated, and the reported accuracies on this benchmark are an artefact of random splitting rather than of the damaged identifier. Both failures are invisible from inside the release: recovering the mapping required matching the images against the source cohort, which a user of the release is not placed to do.

4 Discussion

4.1 Three provenance regimes

The results describe one structure. Patient-wise evaluation is not a property of a splitting algorithm but of the identity evidence the algorithm is given, and that evidence admits three regimes.

In **Regime I** provenance is intact and the graph is equivalent to participant grouping. The null of §3.1 is the design target: a provenance-aware splitter that disagreed with participant grouping on clean data would be wrong. In **Regime II** the participant key is damaged but other relations survive, and the graph recovers part of what was lost: 49% at 10% identifier loss, 90% under full fragmentation. This is the paper's subject. In **Regime III** no relation survives. Tier 1 is there, and so is the right-hand end of the deletion sweep. The limit is informational, not a software limitation, and we claim nothing for the graph in it.

The useful question is therefore not whether a provenance-aware splitter beats participant grouping, but which regime a cohort is in, which can be answered from identifiers alone before any model is trained.

4.2 Why contamination audits disagree about inflation

Vangala *et al.* [15] report that removing every near-duplicate test image leaves balanced accuracy unchanged, and read that as contamination not surfacing as an inflated score. We read their result as compatible with ours, and the difference is in what the two designs can detect.

Their comparison removes duplicated images from the test partition and re-scores a model trained once; no retraining is involved, so the training split still contains those images. A model that has memorised a participant retains that memory after the participant's duplicated images are deleted from the test set. And the identity channel is not confined to near-duplicates: in their own corpus, 28.8% of test images have a near-twin in training while 95.5% of the traceable ones share a *participant* with training. Deleting the first layer leaves the second. Their design measures whether duplicated test items are easier to classify, which is a real question, and not

whether the model learned to recognise participants. Separating those requires an intervention on the training side, which is what the protocol comparison and the permutation control provide. Vangala *et al.* offer three candidate explanations for their null and establish none; the permutation control discriminates between exactly those alternatives.

The practical consequence is narrow and useful. A deduplication audit, the field's common instrument, cannot detect participant-identity leakage, and a benchmark that passes one has not been shown to be sound. Reporting checklists ask whether data were split by patient [28, 29] and risk-of-bias instruments ask whether the split was appropriate [30]; neither asks what identifier the split was performed on, which the measurement here suggests is the question that determines the answer.

4.3 Limitations

The degradation is synthetic, imposed on a cohort with complete provenance rather than observed on one that lost it, which is what makes ground-truth scoring possible and also what makes it a model of the failure rather than the failure itself. A cohort whose damage leaves partial provenance for the graph to carry, Regime II observed rather than induced, is the most informative next experiment. Tier 1 is not that case: its redistribution removed the session and series identifiers along with the participant one.

Replication on an external cohort such as AIBL [31], UK Biobank [32] or OASIS-3 [33] is the immediate next step and is not claimed here. The cohort definitions follow the NIA-AA framework [34, 35], and the architectures are representative of the AD classification literature rather than state-of-the-art [36–39]; nothing in the measurement depends on the backbone, as the DenseNet-121 and volumetric arms show. The public-health framing of early detection [40] is context, not a claim this paper tests.

Every finding is a property of a 2D coronal-centre-slice representation, with a 3D volumetric arm indicating the shortcut survives whole-volume input and is in fact larger there. All intervals are descriptive at five seeds, not significance tests. The contamination dose-response reported in Supplementary Material S4 describes a substitution procedure that also thins the training set and shifts the evaluated class mix, so its slope of +0.125 is not a causal price per unit of leakage. Checkpoint selection, subgroup disparity and the clinical operating point are reported in Supplementary Material S2, S6 and S10 respectively.

5 Conclusions

Patient-wise splitting is not a property of a splitting algorithm. It is a property of the participant identifier the algorithm is given. With identifiers complete the procedure reports 0.771 AUROC on our primary arm; with every identifier removed the same unchanged procedure reports 0.946, against 0.947 for the random image-level splitting it exists to prevent, and no step raises an error anywhere along that path. A study reporting that it split by patient has reported a procedure, not a guarantee.

The endpoints of that axis are not in dispute. What has not been measured is the interval between them, and that interval is where redistributed benchmarks sit: one widely used public benchmark reports 0.999 under the splitting its literature uses and 0.766 once participant identity is recovered. Before trusting a patient-disjoint number, a reader should ask what identifier produced it, and an audit that cannot answer that question has not established soundness.

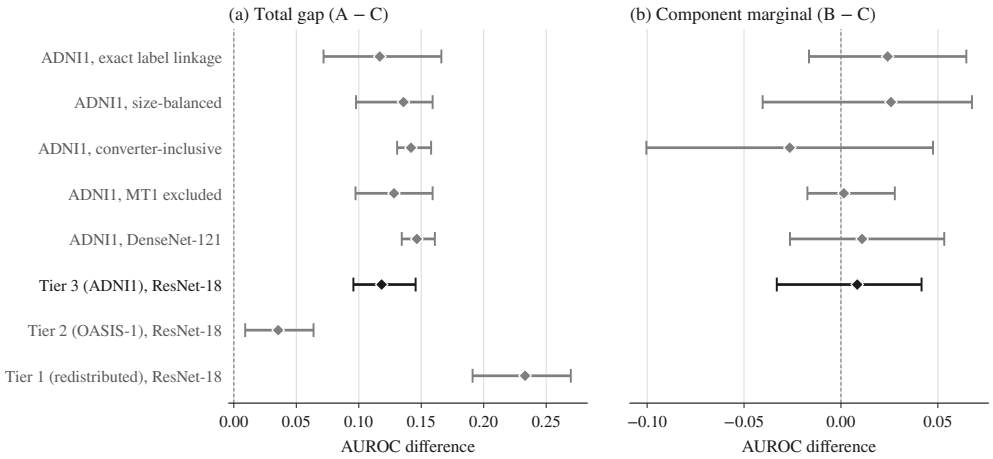


Fig. 1: The two cross-cohort arms and the six ADNI robustness arms, as point estimate and paired-seed bootstrap interval (B = 10,000, n = 5 seeds; descriptive of seed-level variability per the convention in the Results). The primary arm (Tier 3, ResNet-18) is drawn in black. The volumetric arm is reported in Supplementary Material S15 and not drawn here, because it changes architecture, input and training recipe at once and a shared axis would invite the comparison the text declines. **(a)** Total gap, ΔAUROC = Protocol A − Protocol C. The per-protocol AUROCs behind each arm are given in the corresponding subsection; the DenseNet-121 arms reproduce the gap rather than the absolute level (+0.048 on Tier 2 and +0.146 on Tier 3). **(b)** Component-layer marginal, Protocol B−Protocol C, on its own axis because it is an order of magnitude smaller. No interval excludes zero and the converter-inclusive arm inverts; the cross-cohort arms report no marginal and leave the slot empty. The two panels are the paper's headline and its null result on the same runs. The ADNI hierarchical (subject × seed) bootstrap, the more defensible inferential cross-check, gives gap +0.118, CI [+0.061, +0.181], direction preserved.

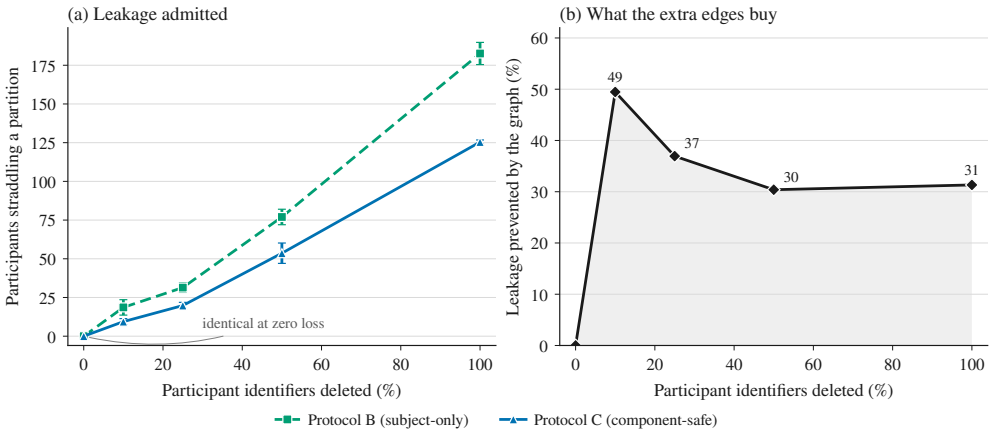


Fig. 2: Provenance degradation on the ADNI1 primary arm, five seeds, mean ± SD. **(a)** Participants whose scans straddle a partition boundary as participant identifiers are deleted. This counts admitted leakage directly, without routing the question through a trained model. The two protocols are indistinguishable at zero loss (the equivalence reported in §2.2, here the origin of a curve rather than a dead end) and separate immediately thereafter. **(b)** The share of that leakage the leakage graph prevents. Both protocols use the same assigner, so the difference is attributable to the grouping; under a mismatched assigner the 10% figure reads 78%. The benefit peaks in the regime where redistributed benchmarks sit and decays once too few identifiers survive for any rule to bind on.

Table 1: Provenance degradation on the ADNI1 primary arm (220 participants, five seeds, mean ± SD). Columns report participants whose scans straddle a partition boundary, and p, the fraction of test-partition participants who also appear in training, the same quantity the dose-response experiment sweeps, so the two curves compose. At zero loss the protocols are identical, which is the §2.2 null result restated as the origin of a curve rather than as a dead end.

Identifiers	**Straddling participants**		p **(test ∩ train)**	
deleted	Protocol B	Protocol C	Protocol B	Protocol C
0%	0.0 ± 0.0	0.0 ± 0.0	0.000	0.000
10%	18.6 ± 5.0	9.4 ± 1.9	0.361	0.207
25%	31.4 ± 3.1	19.8 ± 2.0	0.374	0.408
50%	77.0 ± 5.0	53.6 ± 6.6	0.980	0.807
100%	182.6 ± 7.2	125.4 ± 1.3	0.993	0.990

Table 2: Residual leakage under structured provenance corruption, ADNI1 primary arm (220 participants), five seeds, cell means. Straddling counts true participants with scans on both sides of a boundary; c is the share of test scans whose true participant also appears in training, the quantity Supplementary Material S4 sweeps. Both protocols use the same assigner, so a column difference is attributable to the grouping. MERGE admits no leakage at any intensity, which is why its prevented share is undefined (n/a) rather than zero: over-grouping destroys the participant count without placing a participant on both sides.

Corruption	λ	**Straddling**		**Prevented**	c **(test scans)**	
		Prot. B	Prot. C	by graph	Prot. B	Prot. C
DROP	0	0.0	0.0	n/a	0.000	0.000
	0.1	16.8	8.4	50.0%	0.337	0.177
	0.25	31.2	19.8	36.5%	0.364	0.395
	0.5	85.2	67.0	21.4%	0.991	0.891
	1	182.6	125.4	31.3%	0.989	0.979
SPLIT	0	0.0	0.0	n/a	0.000	0.000
	0.1	1.2	0.6	50.0%	0.007	0.005
	0.25	1.6	0.8	50.0%	0.018	0.004
	0.5	4.0	1.0	75.0%	0.076	0.004
	1	61.6	6.4	89.6%	0.636	0.107
MERGE	0	0.0	0.0	n/a	0.000	0.000
	0.1	0.0	0.0	n/a	0.000	0.000
	0.25	0.0	0.0	n/a	0.000	0.000
	0.5	0.0	0.0	n/a	0.000	0.000
	1	0.0	0.0	n/a	0.000	0.000

Table 3: Every grouping rule the Tier-1 release supports, measured against the recovered participant identities over five seeds. Contamination is the share of test images whose participant also appears in training. The filename key, the only identifier the release provides, leaves seven test images in ten contaminated. Near-duplicate edges at $d \leq 4$ remove an eighth of that; at $d \leq 8$ the graph collapses (8,636 pairs merge the corpus into 27 components, the largest holding 3,117 images, while a further 1,343 firings join opposite-label images and are refused rather than merged), which is a different failure rather than a fix. Only the recovered participant identity yields a clean split.

Grouping rule	Components	Largest (images)	Test contamination mean [min, max]	Test participants in train / total
Filename key (release identifier)	324	32	70.0% [62.0, 78.6]	46.4/59.4
+ near-duplicate $d \leq 4$	219	192	61.2% [53.4, 67.5]	42.0/56.2
+ near-duplicate $d \leq 8$	27	3,117	100%	1.0/1.0
Near-duplicate $d \leq 8$ alone	999	1,770	99.5% [99.0, 100]	104.4/105.0
Recovered participant	200	32	0%	0/29.0

Acknowledgements

Every training run reported here was executed in one code state (approximately 14.8 GPU-hours in total). The software environment of the node that ran the final arms is released in full; the equivalent records for the earlier nodes were not retained. Most arms ran on a single cloud-hosted NVIDIA H100 instance; the 50-run dose-response matrix was rerun in that same code state on an NVIDIA RTX PRO 4500 instance, the 15-run volumetric arm on an NVIDIA RTX 2000 Ada instance, and the 15-run exact-label-linkage arm together with the 120-run structured-corruption matrix, with the 15-run label-permutation control, on an NVIDIA RTX 4090 instance, so those arms and the ones they are compared with differ in hardware but not in code. We report the slope as a descriptive calibration constant and not as a transferable coefficient, which is how it is read throughout. Data used in preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (https://adni.loni.usc.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in the analysis or writing of this report. A complete listing of ADNI investigators can be found at https://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf. OASIS-1 data were provided in part by OASIS Cross-Sectional Principal Investigators: D. Marcus, R. Buckner, J. Csernansky, J. Morris (P50 AG05681, P01 AG03991, P01 AG026276, R01 AG021910, P20 MH071616, U24 RR021382).

Declarations

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics approval and consent to participate

This study used retrospective, de-identified secondary imaging data from the Alzheimer's Disease Neuroimaging Initiative (ADNI) and the Open Access Series of Imaging Studies (OASIS-1), including a public Kaggle benchmark that we show to be a redistribution of OASIS-1 images. No new human participants were recruited, no new imaging was acquired, and no direct participant contact occurred. ADNI and OASIS obtained ethics approval and informed consent at the original data-collection sites; the present secondary analysis was conducted under their respective data-use agreements and public-access terms. The Kaggle benchmark consists of already-public OASIS-1 slice images, and its provenance recovery was performed against OASIS-1 data under the same terms; the recovered image-to-participant mapping is not redistributed. Because this analysis is limited to secondary de-identified data, institutional review board (IRB) approval was not separately sought at Istanbul Medipol University in accordance with local policy on secondary de-identified data reuse. Participant-level

manifests remain local and are hashed via PBKDF2 with a per-release salt so as not to support cross-dataset participant re-identification; only aggregate result tables are redistributed.

Consent for publication

No individual participant data, images, or identifying details are reproduced in this manuscript or in any released artefact, so participant consent for publication is not applicable. All released tables are aggregate, and participant identifiers in the Tier-3 manifests are replaced by a PBKDF2 derivation under a secret salt.

Use of generative AI

Large-language-model coding assistants were used to implement and revise the released software, including the training, audit, split-generation and analysis scripts and the figure and table generators, and to draft manuscript prose. They were also used to locate defects, two of which are reported in the text because they are the kind of silent, self-consistent error this paper concerns. The study design, the protocol definitions, the choice of cohorts and arms and the interpretation of every result are the authors'. No reported number was produced or altered by an LLM: every quantity is computed by the released scripts from the released manifests, and `scripts/verify_paper_numbers.py` re-checks each against the artefact that produced it, so a value introduced into the text would fail that check rather than survive it. The authors reviewed and edited all such output and take full responsibility for the content of this publication.

Author contributions

Mehmet Berke Isler: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Irem Ilter:** Investigation (OASIS-1 cohort), Formal analysis, Writing – review & editing. **Mehmet Kemal Ozdemir:** Supervision, Methodology, Writing – review & editing, Project administration.

Data and code availability

The SplitGuard-AD framework, the training and analysis pipeline and the manuscript sources are released at `https://github.com/mehmetberkeisler/splitguard-ad` under the Apache-2.0 licence, together with the package list frozen on the GPU node (`runs_gpu/gpu_environment.txt`) and `scripts/verify_paper_numbers.py`, which re-checks every number in this manuscript against the artefact that produced it and fails on any disagreement.

The release boundary is aggregate-only and tier-dependent. Scripts, paper sources, figures, aggregate result tables and rebuild instructions are public; raw and preprocessed imaging, model weights, per-image predictions, local manifests and detailed audit rows remain local. `scripts/build_release_artefacts.py` materialises each tier's permitted artefacts into the tracked `release/` directory with a `RELEASE_MANIFEST.json` recording row counts and withheld columns. Tier 1 ships the frozen filename-key split and its component and near-duplicate tables, but not the recovered participant mapping to OASIS-1, which a reader regenerates from their own OASIS-1 copy with `scripts/audit_tier1_ground_truth.py`. Tier 2 ships split manifests indexed by the public OASIS codes, without derived paths or clinical fields. Tier 3 ships hashed component manifests only: participant identifiers are replaced

by PBKDF2-HMAC-SHA256 at 600,000 iterations under a secret per-release salt, because ADNI identifiers are low-entropy and a single-round digest would be invertible by enumeration. ADNI's data use agreement does not permit redistributing participant-level data in any processed form, and model weights trained on it carry the same restriction. Everything withheld is reproducible from the released code plus the data each provider grants directly; Supplementary Material S5 and the repository's `SECURITY_AND_PRIVACY.md` give the full boundary and the checks that enforce it.

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